Fiscal Year:	FY 2009	**Task Last Updated:**	FY 11/05/2008
PI Name:	Lockley, Steven W Ph.D.		
Project Title:	Characteristics of Light Exposure Necessary for Development of Optimal Countermeasures to Facilitate Circadian Adaptation and Enhance Alertness and Cognitive Performance in Space		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Human Factors and Performance Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02115-5804	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 NSBRI-RFA-07-01 Human Health in Space
Start Date:	10/01/2007	**End Date:**	09/30/2011
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Aeschbach, Daniel (Brigham and Women's Hospital) Gooley, Joshua (Brigham and Women's Hospital) Ronda, Joseph (Brigham and Women's Hospital)		
Grant/Contract No.:	NCC 9-58-HPF01301		
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Task Description:	Space missions often expose crew members to unusual light-dark cycles that lead to misalignment between the circadian pacemaker and the sleep-wake schedule, resulting in disturbed sleep and impaired waking function. Sleep disturbances due to other reasons (e.g., noise, temperature, microgravity) further exacerbate cognitive deficits and these factors jointly increase the risk of fatigue-related accidents and injuries. An effective countermeasure is required to facilitate more rapid adaptation of the circadian system and directly enhance alertness and performance. Light exposure induces both of these effects and is an effective, safe, non-pharmacological countermeasure for circadian- and fatigue-related deficits in cognition. Recently, we and others have shown that short-wavelength (blue) light exposure at night is the most effective wavelength for phase-shifting the circadian pacemaker, suppressing pineal melatonin, enhancing subjective alertness, improving performance and inducing EEG-derived brain activation that indicates a more alert state. These effects, however, have only been examined during the night using relatively bright light and recent evidence suggests that the spectral sensitivity of the circadian photoreception system may depend on the intensity and duration of light and that there may be changes in spectral sensitivity during the day. While blue light therapy to facilitate circadian adaptation and alertness has been proposed based largely on the effects of melatonin suppression at night, these data may not be transferable to the effects at other times of day (when melatonin is not produced) or for other effects of light. Before blue light is operationalized, three fundamental questions remain to be determined: 1) Does the spectral sensitivity depend on the intensity of light? 2) Do all non-visual effects of light have the same spectral sensitivity? 3) Is the spectral sensitivity of the circadian photoreception system different between the day and the night? Failure to address these questions may result in deployment of suboptimal or less energy-efficient lighting schedules. The proposed project will test whether the spectral sensitivity of the circadian photoreception system is intensity-dependent, response-dependent or time-of-day dependent, by comparing the effects of exposure to monochromatic light during the day and the night, at both high and low intensities, on circadian phase resetting and enhancement of alertness and performance. Experiment 1 aims to compare the melatonin suppression, phase resetting and alerting effects of exposure to dim 460 nm and 555 nm monochromatic light exposure at night, and compare these data to previously collected findings for bright light exposure at night using the same protocol. Data collection is ongoing and is due for completion within the coming year. There are no results to report at this time. Experiments 2 and 3, which will test the effects of exposure to bright and dim 460 nm and 555 nm monochromatic light during the daytime, will start after completion of experiment 1.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	An effective, safe, well-tolerated, non-invasive, non-pharmacological countermeasure for circadian- and fatigue-related deficits in cognition is required for use during in space environments (space flight and semi-permanent bases) to enhance the safety of crew members. Light exposure has the potential to fulfill this role. Although blue light exposure at night has been shown to be most effective at shifting the circadian pacemaker, and improving alertness and performance, these results may not apply to lower intensities of light (that are routinely experienced during space light) or for light exposure during the biological day. Before light therapy as fatigue countermeasure is operationalized, further research is required to fully understand the intensity- and temporal-dependent characteristics of the circadian photoreception system. The current study will investigate these characteristics in a controlled environment with the long-term view to developing specialized lighting schedules to maximize alertness and performance of space crew in an environment where a even a small fatigue-related errors could have catastrophic consequences.
Task Progress:	We have proposed to complete studies of 12 subjects per year. In our first year, we have initiated studies in 9 subjects to date (4F, age range 18 - 30 years), eight of whom completed the 9-day inpatient protocol. Two more subjects are due to begin the inpatient study in September 2008 and we anticipate reaching and sustaining our proposed recruitment rate (1 subject/month on average) by the end of 2008.
Bibliography Type:	Description: (Last Updated: 11/14/2025)
Articles in Peer-reviewed Journals	Pechacek CS, Andersen M, Lockley SW. "Preliminary method for prospective analysis of the circadian efficacy of (Day)light with applications to healthcare architecture." LEUKOS: The Journal of the Illuminating Engineering Society. 2008 Jul;5(1):1-26. http://dx.doi.org/10.1080/15502724.2008.10747625 , Jul-2008
Articles in Peer-reviewed Journals	Zaidi FH, Hull JT, Peirson SN, Wulff K, Aeschbach D, Gooley JJ, Brainard GC, Gregory-Evans K, Rizzo JF 3rd, Czeisler CA, Foster RG, Moseley MJ, Lockley SW. "Short-wavelength light sensitivity of circadian, pupillary, and visual awareness in humans lacking an outer retina." Curr Biol. 2007 Dec 18;17(24):2122-8. http://dx.doi.org/10.1016/j.cub.2007.11.034 ; PMID: 18082405 , Dec-2007
Articles in Peer-reviewed Journals	Lockley SW. "Spectral sensitivity of circadian, neuroendocrine and neurobehavioral effects of light." Journal of the Human-Environment System. 2008;11(1):43-9. http://dx.doi.org/10.1618/jhes.11.43 , Jan-2008
Awards	Lockley SW. "Invited Speaker, Gordon Conference on Pineal Cell Biology, April 2008." Apr-2008
Awards	Lockley SW. "Keynote speaker, 2nd International Symposium on the Design of Artificial Environments, Fukuoka, Japan, November 2007." Nov-2007
Awards	Lockley SW. "Sleep Splinter Group Co-Lead, Behavioral Health and Performance Meeting, NASA, August 2008." Aug-2008
Awards	Lockley SW. "Symposium Chair, `Therapeutic role of new melatonin receptor agonists', FASEB Summer Research Conference `Melatonin receptors: Actions and therapeutics', August 2008." Aug-2008
Awards	Lockley SW. "Symposium Co-Chair, `Effect of gravity on biological rhythms and sleep', 29th Annual International Gravitational Physiology meeting, Angers, France, June 2008." Jun-2008
Papers from Meeting Proceedings	Lockley SW. "Spectral sensitivity of circadian, neuroendocrine and neurobehavioral effects of light." 2nd International Symposium on the Design of Artificial Environments, Fukuoka, Japan, November 29-December 1, 2007. 2nd International Symposium on the Design of Artificial Environments, Fukuoka, Japan, November 29-December 1, 2007. , Dec-2007

Papers from Meeting Proceedings	Pechacek CS, Andersen M, Lockley SW. "Combining annual daylight simulation with photobiology data to assess the relative circadian efficacy of interior spaces." Proceedings of eSim 2008; Bi-annual conference of the International Building Performance Simulation Association (IBPSA), Quebec City, Canada, May 21-22, 2008. Proceedings of eSim 2008, May 2008. , May-2008